

Learning how to Prove Workshop Week 7

Groups



Cyclic groups

1. Let $G = \langle g \rangle$ and $H = \langle h \rangle$ be cyclic groups, with $|G| = o(g) = m$ and $|H| = o(h) = n$. Suppose that $\gcd(m, n) = 1$.
 - (a) Prove that $o(g, h) = mn$.
 - (b) Conclude that $G \times H$ is cyclic.

2. Let $G = \langle g \rangle$ be a cyclic group of order n . Consider $A = \langle g^a \rangle$ and $B = \langle g^b \rangle$ two subgroups of G . Prove that $A \subseteq B$ if and only if $a \equiv qb \pmod{n}$ for some $q \in \mathbb{Z}$.